

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light switched off

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas-discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

INFORMATION

The stepper motor/servo motor for the dynamic cornering light and its mechanical components and wiring cannot be replaced individually.
The headlight housing must be replaced if repairs are needed.

Diagnostic conditions

- Internal function test on headlight (Xenon) control unit
- Ignition on

Possible fault causes

- ◆ Mechanism of stepper motor for dyn. cornering light sluggish or blocked
- ◆ Dynamic cornering light wiring faulty
- ◆ Stepper motor for dynamic cornering light faulty
- ◆ Headlight control unit faulty

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light, position sensor

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Diagnostic information - Headlight control unit

Stepper motor, dyn. cornering light, version conflict

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INFORMATION

- ♦ Software for the stepper motor does not match software for the control unit; software versions are not compatible.
- ♦ The stepper motor/servo motor for the dynamic cornering light and its mechanical components and wiring cannot be replaced individually.
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Possible fault causes

- ♦ Software version of stepper motor is not supported by control unit
- ♦ Stepper motor for dynamic cornering light faulty
- ♦ Headlight control unit has wrong software version
- ♦ Headlight control unit faulty

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Stepper motor for dyn. cornering light, position sensor

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Stepper motor for dyn. cornering light, position sensor

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Stepper motor for dyn. cornering light, position sensor

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Stepper motor for dyn. cornering light, position sensor

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Diagnostic information - Headlights

Control module faulty

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Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Control module** Headlights faulty

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

Diagnostic information - Headlights

Control module Power supply

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Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Control module** Headlights **Power supply**
- Signal Terminal 15
- Signal Terminal 56b

Fault effects

- ◆ System switches off, Headlights **OFF**
- ◆ Display in instrument cluster

Diagnostic information - Headlight control unit

Control unit, coded incorrectly



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Diagnostic conditions

- Headlight control unit coded incorrectly
- Ignition on

Possible fault causes

- ◆ Line connection from **headlight** control unit to headlight housing connector faulty
- ◆ Wiring at headlight control unit connector Pin 5 and Pin 6 interchanged
- ◆ Headlight control unit faulty

Diagnostic information - Headlights

Adaptive Frontlighting System

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NOTE

Among other things, the **Adaptive Frontlighting System** (AFS) is a dynamic light distribution and is produced by a cylinder that can rotate. It possesses three different positions/contours to change the light/dark limit for ideal highway, motorway and poor-weather lighting.

Diagnostic conditions

- Internal function test on Headlights control unit
- Ignition on
- Driving lights switched on

Fault setting condition(s)

INFORMATION

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- Leads Contacts
- **Servo motor Adaptive Frontlighting System (Headlight housing)**

Fault effects

- ♦ **Adaptive Frontlighting System** Function not available or available only to a limited extent

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Fault setting condition(s)

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◆ The stepper motor/servo motor for the adaptive frontlighting system (AFS) and its mechanical components and wiring cannot be replaced individually.
The headlight housing must be replaced if repairs are needed.

- Leads Contacts
- **Servo motor Adaptive Frontlighting System (Headlight housing)**

Fault effects

- ◆ **Adaptive Frontlighting System** Function not available or available only to a limited extent

Diagnostic information - Headlights

Adaptive Frontlighting System

WARNING

HBA Xenon: Electric shock and burns from high voltage.

- ♦ The lights should always be switched off before removing and installing the headlights.
- ♦ Only operate the gas-discharge lamp when installed in the headlight.
- ♦ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

NOTE

Among other things, the **Adaptive Frontlighting System** (AFS) is a dynamic light distribution and is produced by a cylinder that can rotate. It possesses three different positions/contours to change the light/dark limit for ideal highway, motorway and poor-weather lighting.

Diagnostic conditions

- Internal function test on Headlights control unit
- Ignition on
- Driving lights switched on

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Diagnosis information - (Xenon) gas discharge igniter

Control module Power supply

WARNING

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- ♦ The lights should always be switched off before removing and installing the headlights.
- ♦ Only operate the gas-discharge lamp when installed in the headlight.
- ♦ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

NOTE

NOTE

Sporadic faults!

inactive Fault - Erase fault memory and read it out

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- Lines/connections closed Light bulb Faulty
- Signal Terminal 15
- Signal Terminal 56b

Fault effects

- ♦ System switches off, Headlights **OFF**
- ♦ Display in instrument cluster

Diagnostic information - Headlights

Coding plug Wrong connector assignment

WARNING

HBA Xenon: Electric shock and burns from high voltage.

- ◆ The lights should always be switched off before removing and installing the headlights.
- ◆ Only operate the gas-discharge lamp when installed in the headlight.
- ◆ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

Incorrect coding (pin assignment/contacts) of the headlight control unit connector could be the cause.

- **Control module** Headlights **Pin assignment** incorrect
- **Control module** Headlights **Pin assignment** interchanged

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

Gas discharge lamp, xenon, power supply

WARNING

HBA Xenon: Electric shock and burns from high voltage!

- ◆ Wait an appropriate amount of time for cooling
- ◆ Wear suitable protective gloves to protect your hands.

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or the wiring when the part is removed and disconnected from the power supply.

Diagnostic conditions

- Headlight (xenon) control unit internal function test
- Ignition on

Possible fault causes

- ◆ Wiring harness in headlight faulty (headlight housing)
- ◆ Headlight control unit faulty